

Logistic Regression

Prof. Surya Prakash
IIT Indore

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Recap of Linear Regression

Linear Regression

- Linear regression is a **supervised machine learning algorithm** used for **prediction**

- **Examples**

- **1. Predicting House Prices**

Input variables:	Output:
Area (sq ft)	Price of the house
Number of bedrooms	
Location score	

- **2. Predicting Sales Based on Spending in Advertising**

Input variables:	Output:
TV advertisement budget	Expected sales (in units or revenue)
Radio advertisement budget	
Social media spend	

Linear Regression

- It models relationship between a **dependent variable (output)** and one or more **independent variables (inputs)**.
- Once relationship is established, we can use it for prediction

Input variables: Area (sq ft) Number of bedrooms Location score	Output: Price of the house
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- **Key Idea**

- Linear regression tries to **fit a straight line (in 1D)** or
- A **hyperplane (in multiple dimensions)** that best describes how the output changes with the input.

Linear Regression - Types

- **Simple Linear Regression**

- One input (independent variable)
- Model:

$$y = mx + c$$

or $y = b_1x_1 + b_0$

or $y = b_0 + b_1x_1$

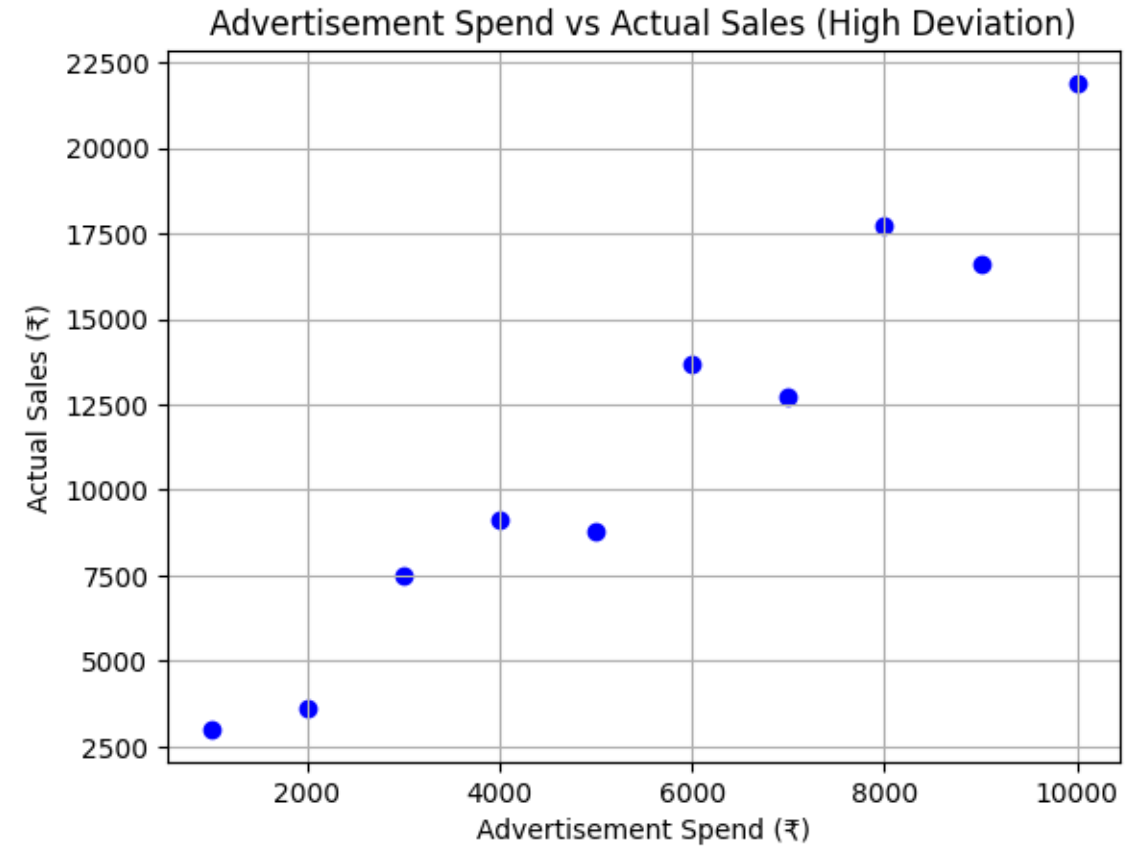
- **Multiple Linear Regression**

- More than one input
- Model:

$$y = b_0 + b_1x_1 + b_2x_2 + \cdots + b_nx_n$$

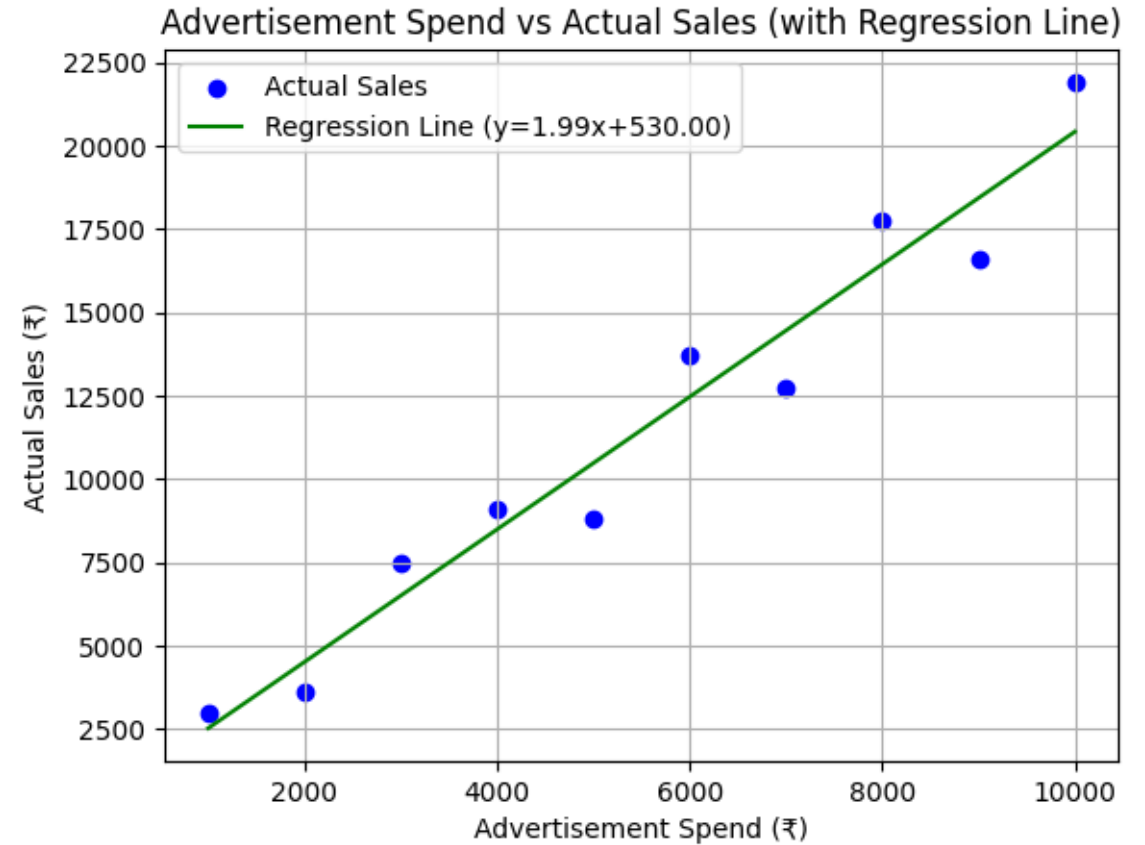
Example - Advertisement vs. Sales dataset

Advertisement Spend (X, ₹)	Actual Sales (Y, ₹)
1000	3000
2000	3600
3000	7500
4000	9100
5000	8800
6000	13700
7000	12700
8000	17750
9000	16600
10000	21900
11000	?
12000	?



Example - Advertisement vs. Sales dataset

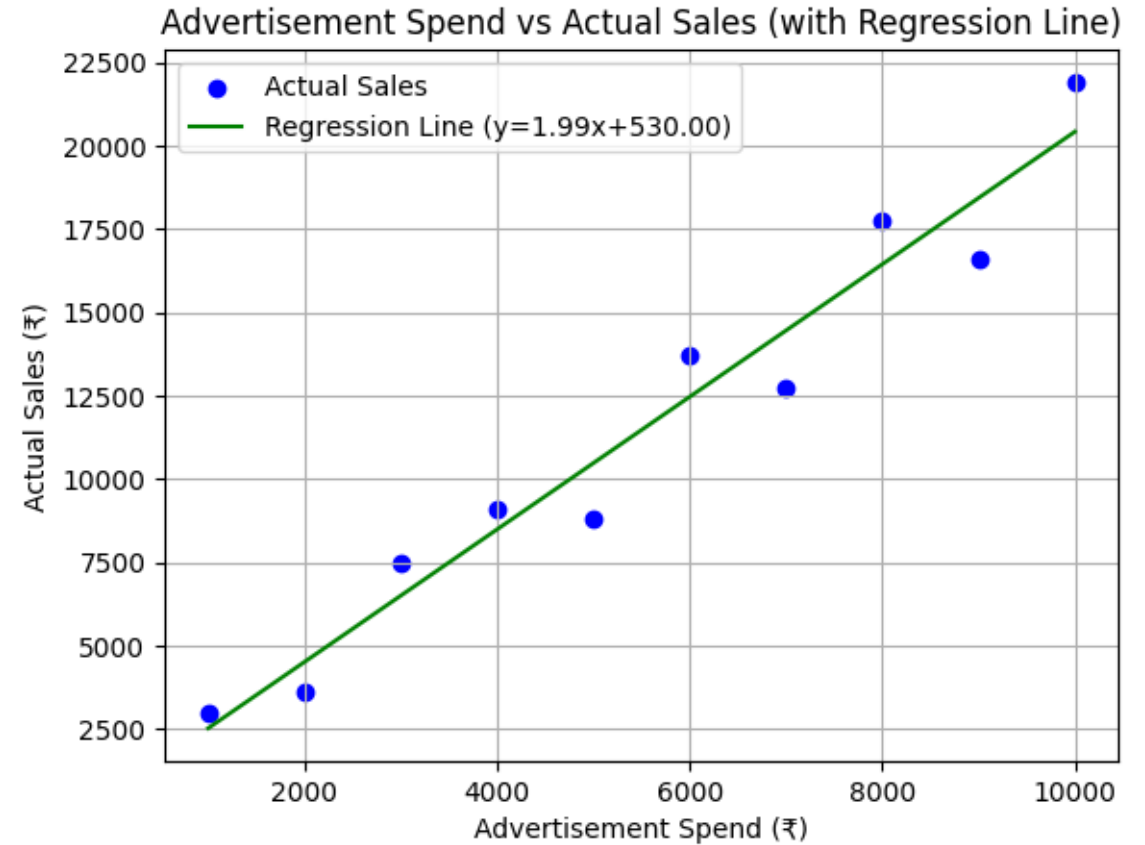
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12000	?



Regression line: $y = 1.99x + 530$

Example - Advertisement vs. Sales dataset

Advertisement Spend (X, ₹)	Actual Sales (Y, ₹)
1000	3000
2000	3600
3000	7500
4000	9100
5000	8800
6000	13700
7000	12700
8000	17750
9000	16600
10000	21900
11000	$y=1.99(11000)+530 = 22420$
12000	$y=1.99(12000)+530 = 24410$



Regression line: $y = 1.99x + 530$

Let's take an example of
loan application

Loan Applications – Scenario 1

- We want to predict the **loan amount** (₹) a person may get based on his/her monthly income.

X → Monthly Income (₹)	Y → Loan Amount Approved (₹)
10,000	50,000
20,000	120,000
30,000	200,000
40,000	280,000
50,000	360,000
60,000	450,000

Model: $Y = \beta_0 + \beta_1 X$

$$Y = -36666.67 + 8X$$

→ Example: If someone earns **₹35,000/month**, the model might predict a loan of ₹2,43,333.

→ Continuous value → **Linear Regression**

Loan Applications – Scenario 2

- We want to predict whether a **loan will be approved (Yes/No)** based on monthly income.
 - Categorical outcome (Yes/No)
- Other similar problems:
 - A given an email is a Spam and Not-Spam
 - Categorical outcome (Yes/No)
 - Whether it will rain tomorrow or not
 - Categorical outcome (Yes/No)

Limitations of Linear Regression

- Linear Regression solves Regression problem and produces continuous output
- Problems mentioned in the previous slide are of Classification
 - Categorical output
- Linear Regression can not solve them
- We need a classifier
 - Logistic regression is required which is a classifier and produces categorical output

Limitations of Linear Regression

- Linear Regression → Solves Regression Problem
 - → Continuous output
- Logistic Regression → Solves Classification Problem
 - Categorical output

Input in both the cases can be continuous or categorical (discrete)

What is logistic Regression?

- **Logistic Regression** is a **supervised machine learning algorithm** used for **classification**, not regression (despite its name).
- It is **mainly** used when the **output (target variable)** is **binary** for example:
 - Will the amount be sanctioned?
 - Is the email spam or not spam?
 - Will the student pass or fail?
- It can be also used for multinomial cases (use of soft-max)

What is logistic Regression?

- Logistic regression produces output in terms of probability
- Subsequently, probability values are converted to class labels using thresholding
- Thresholding for Classification: to make a decision, a threshold is applied (typically 0.5):
 - If $P(y=1) > 0.5$, predict Class 1.
 - If $P(y=1) \leq 0.5$, predict Class 0.

Let's understand Logistic
Regression with the help of
example

Consider Loan Application Scenario 2

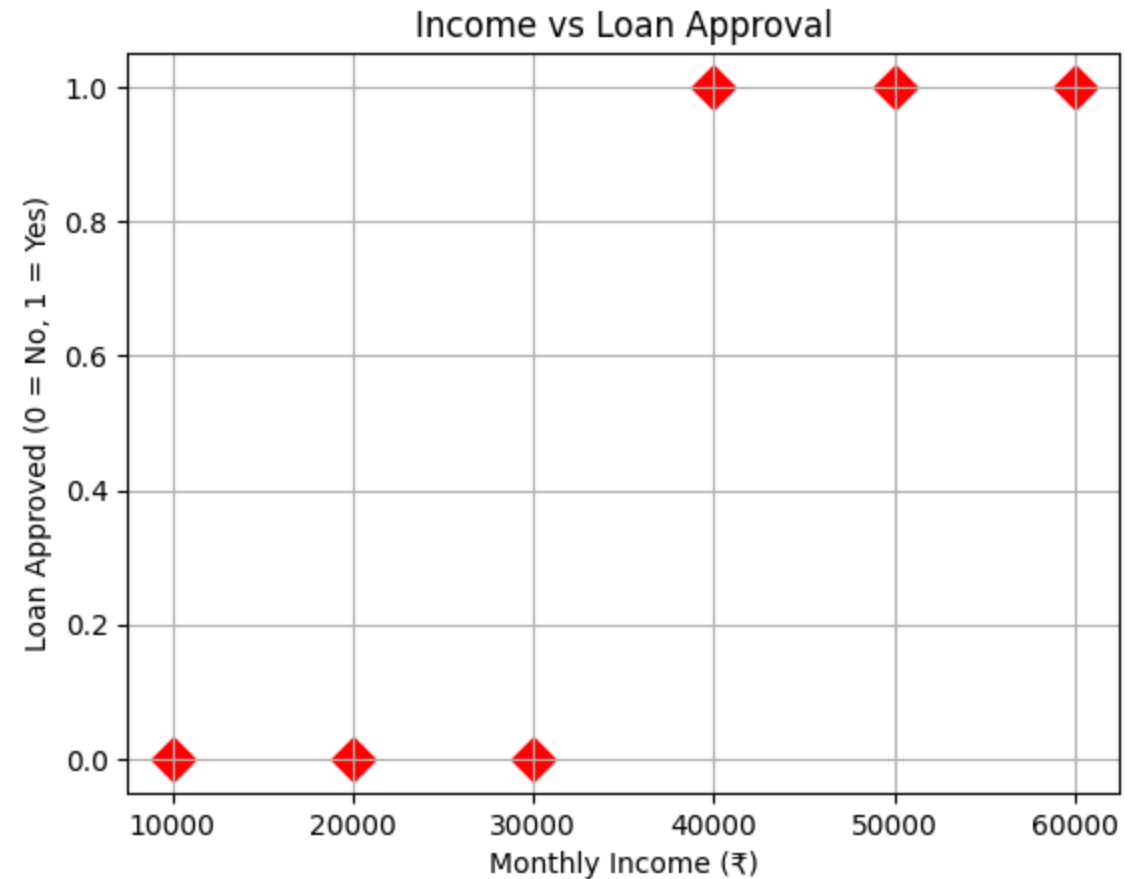
- We want to predict whether a **loan will be approved (Yes/No)** based on monthly income.
- Collect some training data

X → Monthly Income (₹)	Y → Loan Approved? (1 = Yes, 0 = No)
10,000	0
20,000	0
30,000	0
40,000	1
50,000	1
60,000	1

Loan Applications – Scenario 2

- Plot the training data

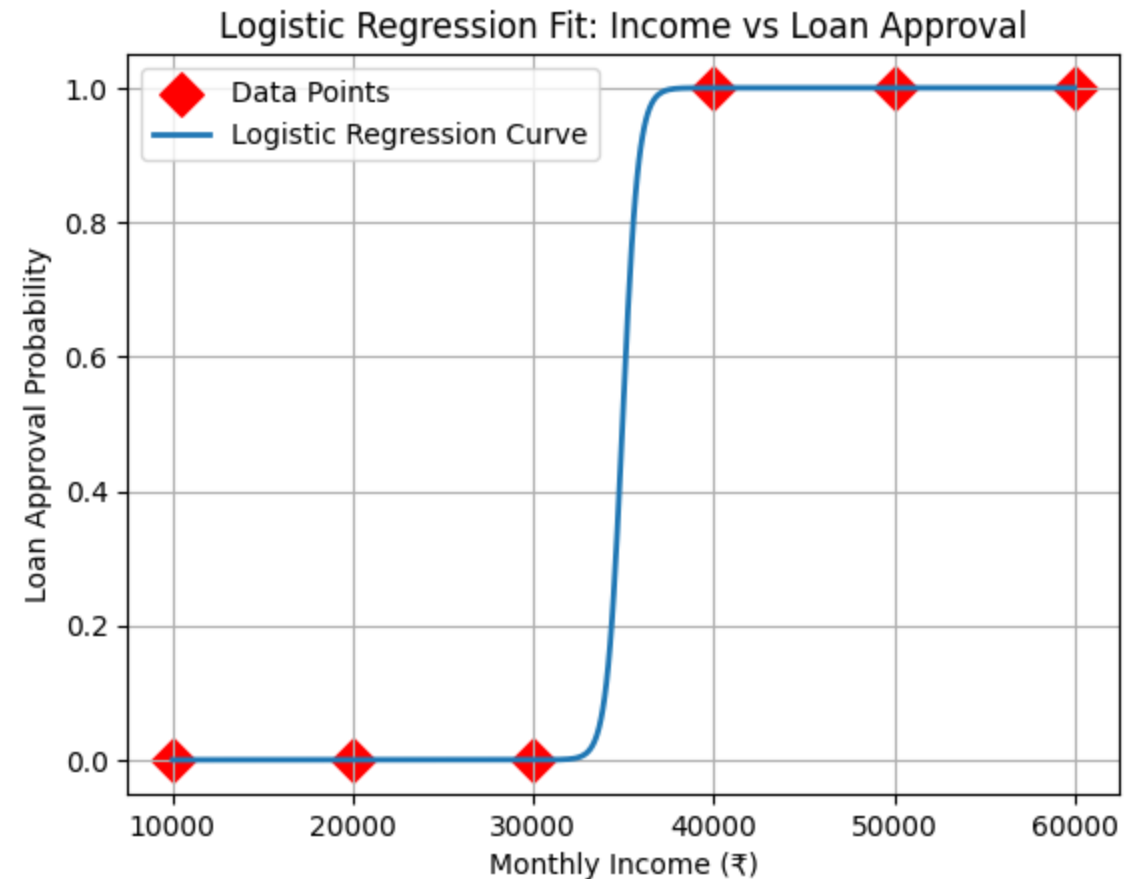
X → Monthly Income (₹)	Y → Loan Approved? (1 = Yes, 0 = No)
10,000	0
20,000	0
30,000	0
40,000	1
50,000	1
60,000	1



Loan Applications – Scenario 2

- Fit the regression curve

X → Monthly Income (₹)	Y → Loan Approved? (1 = Yes, 0 = No)
10,000	0
20,000	0
30,000	0
40,000	1
50,000	1
60,000	1



Loan Applications – Scenario 2

- Objective is to find best fit curve
- Use of S-curve
 - Sigmoid curve (a variant of Logistic curve)

- **Sigmoid Function**

$$P(z) = \frac{1}{1 + e^{-z}}$$

Where:

$$z = w_0 + w_1x_1 + w_2x_2 + \cdots + w_nx_n$$

Loan Applications – Scenario 2

- We want to predict whether a **loan will be approved (Yes/No)** based on monthly income.

X → Monthly Income (₹)	Y → Loan Approved? (1 = Yes, 0 = No)
10,000	0
20,000	0
30,000	0
40,000	1
50,000	1
60,000	1

Model

$$P(Y = 1 | X) = \frac{1}{1 + e^{-(w_0 + w_1 X)}}$$

$$w_0 = -104.7815$$

$$w_1 = 0.0029977$$

Loan Applications – Scenario 2

$X \rightarrow$ Monthly Income (₹)	$Y \rightarrow$ Loan Approved? (1 = Yes, 0 = No)
10,000	0
20,000	0
30,000	0
40,000	1
50,000	1
60,000	1

X = Monthly Income (₹)

Y = Loan Approval (1 = Approved, 0 = Not Approved)

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-(-104.78 + 0.002998X)}}$$

Example: For $X = 35,000$,

$$P(Y = 1 \mid X = 35000) \approx 0.535$$

Loan Applications – Scenario 2

X → Monthly Income (₹)	Y → Loan Approved? (1 = Yes, 0 = No)
10,000	0
20,000	0
30,000	0
40,000	1
50,000	1
60,000	1

Example: For $X = 35,000$,

$$P(Y = 1 \mid X = 35000) \approx 0.535$$

So, the model predicts about **53.5% probability of approval**.

Steps to compute logistic regression coefficients

- Computation of coefficients w_0 and w_1 done using maximum likelihood estimation (MLE),

- **Step 0: Sample Data**

X → Monthly Income (₹)	Y → Loan Approved? (1 = Yes, 0 = No)
10,000	0
20,000	0
30,000	0
40,000	1
50,000	1
60,000	1

Steps to compute logistic regression coefficients

Step 1: Logistic Regression Model

The logistic regression model is:

$$p_i = P(Y_i = 1 \mid X_i) = \frac{1}{1 + e^{-(w_0 + w_1 X_i)}}$$

- When
 - $Y_i = 1, p_i = \text{high}$
 - $Y_i = 0, p_i = \text{low, or } (1 - p_i) = \text{high}$

Steps to compute logistic regression coefficients

- **Step 2: Likelihood Function**
- The **likelihood** for **all observations** is:

$$L(w_0, w_1) = \prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1-y_i}$$

where:

- $y_i = 1$ if Loan Approved = **Yes**, otherwise 0
- $p_i = P(Y_i = 1 | X_i)$, the probability that the i -th observation belongs to Class 1

- **Log-Likelihood (easier to compute)**

$$\ell(w_0, w_1) = \sum_{i=1}^n [y_i \ln(p_i) + (1 - y_i) \ln(1 - p_i)]$$

Steps to compute logistic regression coefficients

- **Step 3: Express p_i in terms of w_0 and w_1**

- The predicted probability is: $p_i = \frac{1}{1 + e^{-(w_0 + w_1 X_i)}}$

- So the **log-likelihood** becomes:

$$\ell(w_0, w_1) = \sum_{i=1}^n \left[y_i \ln \left(\frac{1}{1 + e^{-(w_0 + w_1 X_i)}} \right) + (1 - y_i) \ln \left(\frac{e^{-(w_0 + w_1 X_i)}}{1 + e^{-(w_0 + w_1 X_i)}} \right) \right]$$

- **Simplify:**

$$\ell(w_0, w_1) = \sum_{i=1}^n [y_i(w_0 + w_1 X_i) - \ln (1 + e^{w_0 + w_1 X_i})]$$

Steps to compute logistic regression coefficients

- **Step 4: Maximum Likelihood Estimation**
- To find the values of w_0 and w_1 which maximize the **log-likelihood**, we set:

$$\frac{\partial \ell}{\partial w_0} = 0, \quad \frac{\partial \ell}{\partial w_1} = 0$$

- **For w_0 :**

$$\sum_{i=1}^n (y_i - p_i) = 0$$

- **For w_1 :**

$$\sum_{i=1}^n X_i (y_i - p_i) = 0$$

Steps to compute logistic regression coefficients

- The equations are

$$\sum_{i=1}^n (y_i - p_i) = 0,$$

$$\sum_{i=1}^n X_i (y_i - p_i) = 0,$$

- where

$$p_i = \frac{1}{1 + e^{-(w_0 + w_1 X_i)}}.$$

- Substituting p_i into the equations gives

$$\sum_{i=1}^n \left[y_i - \frac{1}{1 + e^{-(w_0 + w_1 X_i)}} \right] = 0,$$

$$\sum_{i=1}^n X_i \left[y_i - \frac{1}{1 + e^{-(w_0 + w_1 X_i)}} \right] = 0.$$

Steps to compute logistic regression coefficients

- We have

$$\sum_{i=1}^n \left[y_i - \frac{1}{1 + e^{-(w_0 + w_1 X_i)}} \right] = 0,$$

$$\sum_{i=1}^n X_i \left[y_i - \frac{1}{1 + e^{-(w_0 + w_1 X_i)}} \right] = 0.$$

- These are non-linear equations.
- We cannot solve these equations analytically to obtain closed-form expressions for w_0 and w_1 (as we got in Linear regression).

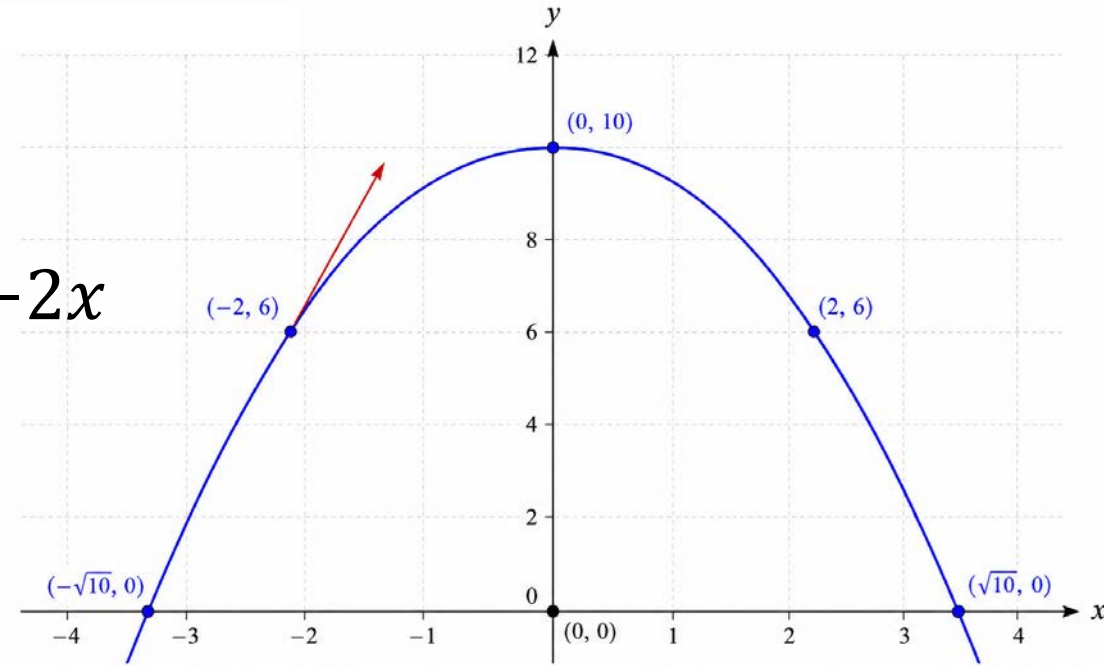
Steps to compute logistic regression coefficients

- So, we use **iterative numerical optimization algorithms**, such as **Gradient Ascent** algorithm to get the values for w_0 and w_1

Gradient Ascent Algorithm – Simple example

- Consider the parabola: $y = 10 - x^2$
 - Compute **max** value of y
- Compute derivative (gradient) $\frac{dy}{dx} = -2x$
- **Apply, gradient ascent**
- The general update rule to reach at x where y gets max value

$$x = x + \eta \frac{dy}{dx}$$



Gradient Ascent Algorithm – Simple example

- Say, $\eta = 0.1$, $x = x + 0.1(-2x)$

- **Iteration 0:**

- Initial point: $x = -2$
- At $x = -2$, $y = 10 - (-2)(-2) = 6$

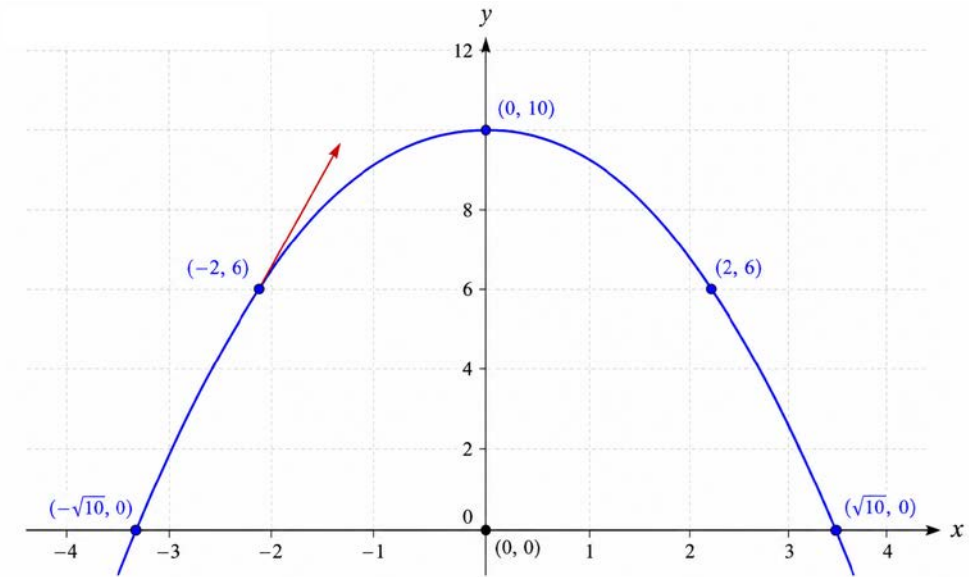
- **Iteration 1**

- Gradient: $\frac{dy}{dx} = -2(-2) = 4$.
- Update: $x = x + 0.1(-2x)$
 $= -2 + 0.1(4) = -1.6$.
- At $x = -2$, $y = 10 - (-1.6)^2 = 10 - 2.56 = 7.44$.
- Since $7.44 > 6$, the algorithm has moved **up the parabola**, toward the maximum.

- **Iteration 2:**

- **Iteration 3:**

- This continues until we get maxima.



Gradient Ascent Algorithm – Simple example

$$\eta = 0.1, \quad x = x + 0.1(-2x)$$

Iteration	x	$y = 10 - x^2$
0	-2.000	6.000
1	-1.600	7.440
2	-1.280	8.362
3	-1.024	8.951
4	-0.819	9.329
5	-0.655	9.571
...	...	...
Converges to	0	10 (maximum)

Gradient Ascent Algorithm – Simple example

Gradient Ascent Algorithm

- Choose initial values (e.g., $w_0 = 0$, $w_1 = 0$).
- Compute the predicted probabilities p_i using $p_i = \frac{1}{1+e^{-(w_0+w_1X_i)}}$
- Compute the gradients:

$$\frac{\partial \ell}{\partial w_0} = \sum_{i=1}^n (y_i - p_i),$$

$$\frac{\partial \ell}{\partial w_1} = \sum_{i=1}^n X_i (y_i - p_i).$$

- Update the parameters:

$$w_0 \leftarrow w_0 + \eta \sum_{i=1}^n (y_i - p_i),$$

$$w_1 \leftarrow w_1 + \eta \sum_{i=1}^n X_i (y_i - p_i),$$

- where η is the learning rate.
- Repeat until the parameter values **converge**.

At convergence, we get the optimum values of w_0 and w_1 which maximizes the **log-likelihood**

End